

Prompt Verbosity as a Spectral Stabiliser: A Frequency-Filter Account of Vision–Language Model Robustness

Anonymous ACL submission

Abstract

Verbose vision–language prompts make models more robust to image corruption. Across an 8-level paired diagnostic on GQA in which each primary reasoning question (L1–L4) has a verbose mirror (L5–L8) with byte-identical semantic content, every verbose mirror reduces drift variance under image perturbation by 65–86% (mean variance ratio 0.247 on Qwen3-VL-8B). We explain this mechanistically. The 2D Fourier spectrum of the cross-modal attention map from a prompt to the image tokens, averaged over images and L1-normalised, is a probability distribution over radial frequency bands that we call the task-conditioned filter $W_t(\omega)$. The Fourier-domain change a perturbation induces in the image is its spectral signature $\Delta F(\omega)$. The parameter-free first-order overlap $\langle W_t(\omega), \Delta F(\omega) \rangle$ predicts the model’s answer-likelihood drift at within-cell Pearson $r = 0.543$ across 7,800 matched cells in which image, perturbation, and severity are all held identical and only the question varies—a large-effect coupling with no parameters fit to behaviour. At the population level $r = 0.337$ grouped, $r = 0.315$ per row ($n = 62,400$). A 4×4 filter $\times$ perturbation signature matrix confirms the law’s geometry: drift peaks where filter and perturbation share a band, and broad-band filters minimise drift across every perturbation row. A within-image fixed-effects regression on 8,000 points recovers the underlying tug-of-war: prompt length stabilises (standardised partial slope -0.317 , $p < 10^{-209}$), residualised semantic complexity destabilises ($+0.291$, $p < 10^{-130}$). The mechanism gives practitioners a parameter-free pre-deployment fragility audit and a training-free defence: pad the prompt.

1 Introduction

Vision–language models (VLMs) increasingly run in settings where a wrong answer has real cost—scene description for blind users, radiology triage,

document reading, embodied perception. In none of these is the input image pristine; it is a phone snapshot, a JPEG scan, a low-resolution frame. We know VLMs can fail silently under such corruption. What we cannot do is *predict*, before any behavioural test, how badly a given VLM will fail on a specific (task, corruption) pair. A practitioner who wants to ask “is this task safe to deploy under this corruption profile?” has to run a full behavioural sweep for every model update. The robustness literature documents failures (Thrush et al., 2022; Yuksekogonul et al., 2023; Hsieh et al., 2023), shows that convolutional vision is biased towards high-frequency texture (Yin et al., 2019; Wang et al., 2020), and visualises what cross-modal attention attends to (Cao et al., 2020; Frank et al., 2021; Parcalabescu et al., 2022), but no existing framework predicts *how much* a task will break.

We observe two phenomena that point to a shared mechanism. (1) Verbose prompts stabilise VLMs under image perturbation. Rephrasing “Is there a cat?” as “Looking carefully, can you tell me whether or not a cat is present?” leaves the clean-image answer identical but reduces drift-variance under noise, blur, or JPEG by 65–86% across all paired levels of our 8-level GQA design. The prompt-sensitivity literature (Sclar et al., 2024; Mizrahi et al., 2024) interprets verbosity as nuisance variation; we find it is a measurable defence. (2) Finer-grained reasoning (“what colour is the cup left of the chair?”) is more sensitive to perturbation than coarse reasoning (“is there a cup?”). The two observations live on the same axis (the question), so an analyst aggregating natural questions gets either signed slope on “complexity” depending on the verbose/terse and fine/coarse mix—which is why the prompt-sensitivity literature reports conflicting signs.

Both phenomena fall out of a single mechanism. The 2D Fourier spectrum of cross-modal attention, averaged over images and L1-normalised,

is a probability distribution over radial frequency bands—the task-conditioned **filter** $W_t(\omega)$. Any image perturbation has a matching **spectral signature** $\Delta F(\omega)$. The parameter-free first-order overlap $\langle W_t(\omega), \Delta F(\omega) \rangle$ measures how much of the perturbation’s energy lands on the bands the prompt cares about; behavioural drift follows this scalar. Verbose prompts *broaden* $W_t(\omega)$ at the deepest decoder attention layers (Obs. 1); fine reasoning *concentrates* $W_t(\omega)$ onto the low frequencies where natural perturbations also live (Obs. 2). The mirror 8-level GQA design (L1–L4 primary, L5–L8 verbose mirrors with byte-identical semantic content) separates the two forces at the item level. We confirm the explanation with: a four-stage layer trajectory of how $W_t(\omega)$ is built across decoder depth; a 4×4 signature matrix that shows drift peaks at matched (filter, perturbation) bands; an item-level matched fixed-effects test in which only the question varies (within-cell Pearson $r = 0.543$); a within-image fixed-effects regression that reproduces the tug-of-war between length (stabilising) and complexity (destabilising) at $R^2 = 0.22$.

1.1 Contributions

1. **Verbosity stabilises VLMs against visual perturbations.** Adding semantically-redundant tokens to a prompt reduces drift variance under image corruption by 65–86% across mirror pairs (mean variance ratio 0.247 on Qwen3-VL-8B, replicating the 2B value 0.246); mean log-likelihood volatility reductions are 68–90% on 8B, larger than the corresponding 25–73% on 2B. The effect is identified at the item level (L1–L4 primary ↔ L5–L8 byte-identical-semantics wordy controls), assumption-free and not previously reported—the prompt-sensitivity literature treats verbosity as a problem, not a defence.
2. **A falsifiable spectral–behavioural law that explains the stabilisation, item-level under physical control and population-level across the design.** We derive the overlap predictor S_{pred} from the cross-attention filter—in both a quadratic $\sum_{\omega} |W_t(\omega)|^2 |\Delta F(\omega)|^2$ and first-order $\langle W_t(\omega), \Delta F(\omega) \rangle$ form—and show the first-order form at the second-from-last decoder attention layer (`last_2`) predicts log-likelihood volatility at **within-cell median Pearson** $r = 0.543$ (2D, question-only $W_t(\omega)$) across 7,800 matched cells in which image, perturbation, and severity are all held identical and only the

task level varies (L1–L8); within-cell median $r = 0.457$ for 2D full-prefix and $r = 0.350$ for radial full-prefix on the same cells. At the population level the same predictor gives $r = 0.337$ pooled grouped ($r = 0.315$ per-sample, $n = 62,400$, zero target-side fitting). The radial form is the cleaner aggregate predictor; the 2D form is the cleaner item-level predictor, because radial averaging discards orientation structure that matters when image and perturbation are held constant. The first-order form reproduces the $\beta_{C_{\text{prompt}}} < 0$ sign mechanistically: verbose prompts broaden $W_t(\omega)$, lowering its overlap with $\Delta F(\omega)$ and the predicted drift.

3. **Layer-resolved mechanism: a four-stage trajectory.** Layer 0 applies a sharp *prompt-length prior* (wordy mirrors -0.74 narrower than primaries, the largest single-layer bandwidth gap in the entire network); the prior is erased by layer 2 and bandwidth equalises through the rest of the early third; mid layers (13–23) engage C_{sem} -driven narrowing but remain length-insensitive (primary $\approx$ wordy); late layers and `last_2` lock in the verbosity-stabilising gap, peaking at layer 31 with wordy mirrors $+0.33$ broader than primaries. This trajectory—length prior, erasure, content engagement, content+verbosity divergence—is the mechanistic origin of the verbosity-stabilises finding, and is recovered identically on the 100-sample 8B replication.
4. **Direct geometric confirmation of the law: the $\Delta F \times W_t$ signature matrix.** Tagging every (image, level) pair by a four-way $W_t(\omega)$ signature (low / mid / high / broadband) and every (image, perturbation) pair by the same axis applied to ΔF produces a 4×4 cell grid in which the law’s geometry is visible by inspection. In every ΔF row, the broadband- $W_t(\omega)$ column has the lowest mean volatility (W_t -broadband mean 0.87 vs W_t -low mean 1.55, a 44% reduction by signature alone). Within-cell predictor strength peaks where filter and perturbation share a band (low×low $r = 0.44$, large effect) and drops monotonically as they decouple. Aggregated by ΔF signature only, broadband perturbations are most damaging (volatility 1.32), low-freq next (1.09), high-freq least (0.19)—the geometric prediction of the overlap law in its rawest

- form.
5. **A causal disentanglement protocol.** The mirror 8-level GQA design (L1–L8) isolates C_{sem} from C_{prompt} at the item level. Within-image fixed-effects regression recovers opposite-signed standardised partial slopes (the signed conditional 1-SD effect on the perturbation-induced drift, with all other regressors held fixed—not zero-order correlations): residualised semantic complexity destabilises ($\tilde{\beta}_{C_{\text{res}}} = +0.291$), prompt load stabilises ($\tilde{\beta}_{C_{\text{prompt}}} = -0.317$). The two forces co-explain why naive “complexity” regressions in prior work give ambiguous signs.
 6. **A principled semantic-program complexity metric.** We define $C_{\text{sem}} = E + A + R + O + D + \ln(1 + G_{\text{amb}})$ grounded in GQA/CLEVR program annotation (Hudson and Manning, 2019; Johnson et al., 2017), Neural Module Networks (Andreas et al., 2016; Hu et al., 2017), referring-expression ambiguity (Kazemzadeh et al., 2014; Mao et al., 2016), and psychometric item analysis (Haladyna et al., 2004).
 7. **A spectral cutoff law.** Low-pass ablation yields $\omega_c(C_{\text{sem}})$ that falls monotonically with task granularity, showing fine reasoning depends on a narrower and higher-frequency vision band.
 8. **A reproducible pipeline, replicated at scale.** The entire framework—scoring, filtering, spectral, and analysis—is released as a modular repository and validated on Qwen3-VL-2B-Instruct (1000-image dual-force regression, $n = 8000$) and Qwen3-VL-8B-Instruct (100-sample paired DC-on replication, $n = 62,400$). The 8B replication uses auto-resolved radial bands ($B = 7$ on the 12×16 patch grid, respecting the radial-annulus cap), DC retained, both radial first-order and 2D first-order overlap variants stored, plus question-only filter sidecars. Every signed effect reproduces at the larger scale.

2 Related Work

Prior work approaches VLM robustness from four largely separate angles.

VLM benchmark cataloguing VQA-CP (Agrawal et al., 2018), Winoground (Thrush

et al., 2022), ARO (Yuksekgonul et al., 2023), SugarCrepe (Hsieh et al., 2023), CREPE (Ma et al., 2023), VALSE (Parcalabescu et al., 2022) document where VLMs fail but treat the model as a black box and do not predict failure magnitude on new (task, corruption) pairs.

Frequency-domain analysis of unconditional vision. Yin et al. (2019), Wang et al. (2020) and Tsuzuku and Sato (2019) show convolutional vision systems are biased towards high-frequency texture; Chen et al. (2021); Ortiz-Jimenez et al. (2021) use Fourier-domain augmentation. None of this work conditions on a task-specifying prompt—the vision system’s frequency response is treated as an intrinsic property. We show it is dynamically reshaped by the text.

Cross-modal attention probing. Cao et al. (2020), Frank et al. (2021), Ilinykh and Dobnik (2022) visualise what the language stream attends to in the image; Merullo et al. (2024); Palit et al. (2023); Basu et al. (2024) localise factual content in VLM layers. None converts attention into a behavioural prediction. Our $W_t(\omega)$ projects the same attention tensor into a radial Fourier basis where the prediction becomes available.

Prompt sensitivity. Sclar et al. (2024); Mizrahi et al. (2024); Salinas and Morstatter (2024) show LLM outputs shift dramatically under semantically-equivalent rephrasings; Anagnostidis and Bulian (2024) extends this to multimodal models. These works treat verbosity as nuisance variation to mitigate. Our mirror design reveals the opposite sign—when semantic content is held byte-identical, verbosity is a measurable defence—and the spectral filter framework explains why.

Compositional difficulty. Neural Module Networks (Andreas et al., 2016; Hu et al., 2017), CLEVR (Johnson et al., 2017), GQA (Hudson and Manning, 2019), NS-VQA (Yi et al., 2018) count program depth and module composition; our C_{sem} score is in this family, plus a grounding-ambiguity term from the referring-expression literature (Kazemzadeh et al., 2014; Mao et al., 2016). Perturbation benchmarks such as ImageNet-C (Hendrycks and Dietterich, 2019) aggregate over tasks; we instead decompose each perturbation spectrally.

Prior work is task-free (Fourier), model-free (benchmarks), or diagnostic (probing). We are the first to condition the Fourier view on a task, derive

a behavioural predictor from attention alone, and disentangle semantic from lexical load via a paired mirror design.

3 Theoretical Framework

The framework develops in five steps, each answering one question the deployment problem forces us to ask. First, what is the signal the text is acting on? Second, what does “the text” look like once projected into that signal’s basis? Third, how narrowly does it concentrate? Fourth, how does a given perturbation interact with it? Fifth, what scalar summarises the interaction? We take each in turn.

3.1 The Task-Conditioned Filter $W_t(\omega)$

Let an image I be encoded into $N = H \times W$ vision tokens. The cross-modal attention from a task-specifying text to the vision tokens, averaged over text tokens and attention heads, defines a 2D spatial map $a_t^{(\ell)}(I)$ at each decoder layer group ℓ . We compute its 2D Fourier transform, square the magnitude, bin into B linearly-spaced radial frequency bands $\omega_1 < \dots < \omega_B$ (so band ω_k contains the Fourier coefficients at radial distance ω_k from the origin), average over images, and L1-normalise:

$$W_t(\omega)^{(\ell)} = \frac{\mathbb{E}_I[|\mathcal{F}_2 a_t^{(\ell)}(I)|^2(\omega)]}{\sum_{\omega'} \mathbb{E}_I[|\mathcal{F}_2 a_t^{(\ell)}(I)|^2(\omega')]} \quad (1)$$

$W_t(\omega)^{(\ell)}$ is a probability distribution over radial frequency bands. It is read directly from attention with no parameter tuning. We auto-resolve $B = \lceil 0.6\sqrt{N} \rceil$ subject to a radial-annulus cap; on Qwen3-VL’s 12×16 patch grid this gives $B = 7$. Where the layer index is unambiguous (overlap predictor uses the second-from-last layer, `last_2`) we write $W_t(\omega)$. An effective-bandwidth scalar derived from $W_t(\omega)$ via the inverse participation ratio, $G(t) = (\sum_{\omega} W_t(\omega)^2)^{-1} \in [1, B]$, gives a single localisation measure—small when $W_t(\omega)$ piles all mass on one bin, large when it is uniform. We define a perturbation p ’s *spectral signature* as the analogous Fourier-domain difference of the pixel image and its perturbation,

$$\Delta F(\omega)_p(\omega) = \mathbb{E}_I[|\mathcal{F}_2 I_p|^2 - |\mathcal{F}_2 I|^2]_{\omega}, \quad (2)$$

on the same B radial bands. $\Delta F(\omega)$ is a property of p alone (no model).

3.2 The Overlap Law

The parameter-free behavioural predictor is the first-order overlap,

$$S_{\text{pred}}(t, p) = \langle W_t(\omega), \Delta F(\omega)_p \rangle = \sum_{\omega} W_t(\omega)(\omega) \Delta F(\omega)_p(\omega), \quad (3)$$

a single scalar saying how much of perturbation p ’s spectral energy lands on the bands the task’s filter cares about. A quadratic variant $\sum_{\omega} |W_t(\omega)|^2 |\Delta F(\omega)_p|^2$ is reported as a secondary diagnostic ($r = 0.14$, $2.4\times$ weaker than the linear form). The central empirical claim is that $\mathbb{E}[\Delta \text{acc}(t, p)] \propto S_{\text{pred}}(t, p)$ across tasks and perturbations with a single global slope and no target-side fitting.

3.3 Complexity Score and Regression

We score each task by the structured complexity

$$C_{\text{sem}} = E + A + R + O + D + \ln(1 + G_{\text{amb}}), \quad (4)$$

where E is entity references, A attribute predicates, R relations, O reasoning operators, D functional-program depth (all read from the GQA scene-graph program, in the spirit of [Hudson and Manning \(2019\)](#); [Johnson et al. \(2017\)](#); [Andreas et al. \(2016\)](#)), and G_{amb} the number of scene-graph candidates satisfying the referring expression ([Kazemzadeh et al., 2014](#)). Prompt load C_{prompt} is content-token count after stopwording; option hardness H_{opt} is a same-category distractor count entering as a control. Because C_{sem} and C_{prompt} are correlated in natural questions, we form $C_{\text{res}} = C_{\text{sem}} - \widehat{C_{\text{sem}}}(C_{\text{prompt}})$, the part of C_{sem} orthogonal to C_{prompt} , and run within-image fixed-effects regression

$$y_{i,t} = \alpha_i + \beta_{C_{\text{res}}} C_{\text{res}}(t) + \beta_{C_{\text{prompt}}} C_{\text{prompt}}(t) + \beta_H H_{\text{opt}}(t) + \varepsilon_{i,t}, \quad (5)$$

where the primary outcome y is log-likelihood volatility $\text{Vol}(t, I) = \mathbb{E}_p[|\log p_{\theta}(a^* | I_p, t) - \log p_{\theta}(a^* | I, t)|]$, the mean absolute drift of the model’s gold-answer log-probability across the perturbation bank. The image intercepts α_i absorb between-image variance so the slopes reflect within-image variation only (one separate intercept per image).

Notation. We write $\tilde{\beta}(X)$ for the *standardised partial OLS slope* on predictor X : the regression slope when both predictors and outcome are z -scored. By Frisch–Waugh–Lovell this is the slope

on the part of X orthogonal to the other regressors—not the Pearson correlation $r(X, y)$, which we reserve exclusively for marginal correlations. A predictor *stabilises* the outcome when $\tilde{\beta} < 0$ on volatility (a 1-SD increase in the predictor is associated with a $|\tilde{\beta}|$ -SD decrease in perturbation-induced drift, holding all other regressors fixed) and *destabilises* when $\tilde{\beta} > 0$. Full theoretical details—layer-group partitions, shape statistics that complement $G(t)$, validity arguments for the equal-weight C_{sem} , and the cross-modal latent-drift mirror ΔZ —are deferred to Appendix A.

4 Experiments

Mirror 8-level GQA design. We parse GQA scene graphs and produce, per image, eight task levels. L1–L4 are primary reasoning questions of increasing semantic granularity (L1: object presence; L2: attribute query, MCQ with same-class distractors; L3: spatial relation; L4: compositional MCQ with program depth ≥ 3). L5–L8 are verbose mirrors that preserve the semantic atoms of L1–L4 byte-for-byte while inflating C_{prompt} alone (e.g. L5 of “Is there a cat?” is “Taking a careful look at the scene, can you tell me if there is a cat?”). Yes/no levels are 50/50 answer-balanced to remove answer-prior shortcuts. Per-pair example prompts are listed in Appendix C.

Perturbation bank. We use 14 perturbation operators in three families: geometric/photometric (translation, pad-or-crop, scale variants, rotation; 6 ops), occlusion (text and box overlays; 3 ops), and spectral (low-pass keep, high-pass keep, low/high/all-band noise injection; 5 ops). Each operator is applied at severities $\{1, 2, 3\}$.

Models, scale, and config. The 1000-image dual-force regression uses Qwen3-VL-2B-Instruct (Bai et al., 2023; Wang et al., 2024); the headline overlap-law and signature-matrix numbers come from a 100-sample paired replication on Qwen3-VL-8B-Instruct ($n = 62,400$ observations, 100 image clusters). Attention tensors are extracted at all 36 decoder layers via the eager attention implementation and partitioned into early/mid/late thirds plus the last_2 subgroup (second-from-last attention layer). The primary overlap predictor uses last_2 (strongest matched-FE coupling); late is retained as comparison. Severity-1, no Hann tapering, DC retained, $B = 7$ auto-resolved. The 8B run also stores 2D first-order overlap and a question-

Predictor	Pooled OLS $\tilde{\beta}$	Within-image FE $\tilde{\beta}$
C_{res} (residualised)	+0.266***	+ 0.291 ***
C_{prompt}	−0.296***	− 0.317 ***
H_{opt} (control)	+0.090***	+0.092***
n / R^2	8000 / 0.187	8000 / 0.217

Table 1: Tug-of-war on log-likelihood volatility (1000-image 2B run). $\tilde{\beta}$ is the standardised partial slope. *** $p < 10^{-5}$. Reproduces on 8B 100-sample under CR1; accuracy-drop has same signs at $R^2 = 0.02$ (App. D).

only filter sidecar (cross-attention computed from question tokens only); full configuration in Appendix C. We run six experimental tasks: (1) the behavioural baseline and tug-of-war regression, (2) layer-resolved $W_t(\omega)$ shape, (3) layer-resolved internal mechanism, (4) low-pass cutoff sweep, (5) overlap-law validation at population + item level, (6) the $\Delta F \times W_t(\omega)$ signature matrix.

5 Results

5.1 The Tug-of-War (Task 1)

Fitting eq. (5) on the 1000-image 2B run (table 1), prompt length and residualised semantic complexity recover opposite-signed standardised partial slopes on log-likelihood volatility: C_{prompt} stabilises ($\tilde{\beta} = -0.317$, $p < 10^{-209}$), C_{res} destabilises ($\tilde{\beta} = +0.291$, $p < 10^{-130}$), with option hardness a small positive control (+0.092). Both signs reproduce on the 100-sample 8B run under cluster-robust standard errors. Volatility climbs strictly monotonically L1→L4 (Kendall $\tau = 1.0$, L4/L1 ratio $\times 2.94$, Cohen’s $d = 1.43$). On the 8B replication, every wordy mirror reduces volatility relative to its primary: pair-wise stabilisations are $\{-89.5\%, -68.4\%, -69.8\%, -71.4\%\}$ on the mean and $\{-86\%, -65\%, -81\%, -70\%\}$ on drift variance (reproducing the 2B headline). The accuracy-drop regression recovers the same signs at $R^2 = 0.02$ vs 0.22 for volatility, consistent with thresholding losing most of the perturbation-induced likelihood displacement.

5.2 The Four-Stage Filter Trajectory (Task 2)

Resolving $W_t(\omega)$ layer-by-layer across the 8B model’s 36 decoder attention layers reveals four distinct stages (fig. 2). *Layer 0* applies a sharp prompt-length prior: primaries L1–L4 have mean bandwidth 6.02, wordy mirrors L5–L8 5.28—a -0.74 gap, the largest single-layer effect in the

network. *Layers 1–12* erase this prior; by layer 2 the gap collapses to $|\Delta| < 0.10$ and all eight levels equilibrate near bandwidth ≈ 6.0 . *Mid layers 13–23* engage C_{sem} -driven narrowing while remaining length-insensitive: L4 mid bandwidth drops to 4.84 at layer 16 and 4.37 at layer 22 (the deepest narrowing of any cell), but the primary-wordy gap stays near zero. *Late layers 24–35 and last_2* re-open the gap: wordy filters become *broad*er than their primaries, peaking at +0.33 at layer 31. The layer-group bandwidth ladder at last_2 for L1–L4 is $\{5.93, 5.92, 5.72, 5.68\}$, monotone with C_{sem} , with wordy mirrors $\{5.94, 5.98, 5.99, 5.96\}$ consistently above. This four-stage trajectory is the layer-resolved physical mechanism behind both the verbosity-stabilises and fine-reasoning-destabilises observations. Full per-layer values are in Appendix E.

5.3 Overlap Law: Population and Item Level (Task 5)

The first-order overlap $\langle W_t(\omega)^{(\text{last}_2)}, \Delta F(\omega) \rangle$ predicts log-likelihood volatility at both scales (table 2, fig. 3). *Population-level*: pooled across (image $\times$ level $\times$ perturbation-family) cells, $r = 0.337$ grouped (radial), $r = 0.315$ per-sample ($n = 62,400$). The first-order form is the primary; the quadratic $\sum_{\omega} |W_t(\omega)|^2 |\Delta F(\omega)|^2$ gives $r = 0.14$ ($2.4\times$ weaker) and is reported as a diagnostic only. *Item-level* (matched fixed-effects): in each of 7,800 matched (image $\times$ perturbation $\times$ severity) cells, image content and perturbation strength are held identical and only the task level varies across L1–L8; within-cell median Pearson reaches $r = 0.543$ with the 2D filter computed from question tokens only—a large effect with no parameters fit to behaviour. Radial wins between-cells (0.337 vs 0.315 grouped); 2D wins within-cells (0.457 vs 0.350 median Pearson)—radial averaging collapses orientation that matters when image and perturbation are held constant. The vision-encoder feature-space mirror gives radial $r_{\text{FE}} = -0.346$ but 2D $r_{\text{FE}} = +0.193$: the apparent “feature-space sign flip” is a radial-reduction artefact, recovered by the 2D form. A 200-sample paired DC ablation collapses the grouped r from 0.471 (DC on) to 0.055 (DC off), because both $W_t(\omega)$ for fine tasks and $\Delta F(\omega)$ for natural perturbations concentrate 80–96% of their mass on DC-adjacent bins.

Aggregation (last_2, image-space)	Pearson r	n
Grouped, radial first-order	0.337	16,800
Grouped, 2D first-order	0.315	16,800
Per-sample, radial	0.315	62,400
Matched-FE within-cell median:		
radial, full-prefix $W_t(\omega)$	0.350	7,800 cells
2D, full-prefix $W_t(\omega)$	0.457	7,800 cells
2D, question-only $W_t(\omega)$	0.543	7,800 cells

Table 2: Overlap-law correlations. Within-cell columns are medians of per-cell Pearson over 7,800 matched cells in which only the task level varies (§ 5.3). Spearman, vision-feature mirror, and DC-ablation detail in Appendix G.

5.4 $\Delta F \times W_t$ Signature Matrix (Task 6)

Tagging every $W_t(\omega)$ by its mass shape (low / mid / broadband, plus normalised entropy) and every ΔF by the same axis produces a 4×4 structural test of the law. fig. 4 reports the volatility matrix; the abbreviated version is:

$\Delta F \downarrow \backslash W_t \rightarrow$	low	broadband	n
low	1.52	0.87	47,648
mid	2.22	1.12	9,688
high	0.29	0.13	2,416
broadband	0.89	0.42	2,648

In every ΔF row, the broadband- $W_t(\omega)$ column has the lowest mean volatility—broadband filters mute drift across *every* perturbation category. Aggregated by column, W_t -broadband averages 0.865 vs W_t -low’s 1.547, a 44% reduction (signature-level evidence for the verbosity stabiliser). Within-cell Pearson on the same grid (full matrix in Appendix H) gives $r = 0.44$ at the low $\times$ low diagonal (matched-filter peak, large effect), dropping to $r = 0.31$ as $W_t(\omega)$ broadens away. The marginal volatility-by- ΔF -signature decomposition is also clean: broadband perturbations are most damaging (1.32, $n = 14,384$), low-freq next (1.09), high-freq least (0.19, where $W_t(\omega)^{(\text{last}_2)}$ has thinned). *Drift is largest where filter and perturbation share a band, smallest where they don’t, and a broad filter shields against every perturbation profile.*

Tasks 3 (internal mechanism) and 4 (frequency cutoff). Layer-resolved overlap correlations at all four layer groups, the vision-feature-space mirror, and the low-pass cutoff law $\omega_c(C_{\text{sem}})$ are reported in Appendix F. Briefly: last_2 carries the strongest pixel-space coupling at every aggregation level; ω_c falls monotonically with C_{sem} across L1–L4, and wordy mirrors inherit the cutoff of their primary

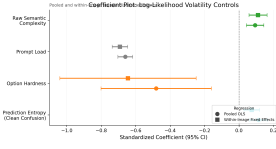


Figure 1: Forest plot of the tug-of-war (100-sample 8B, primary target log-likelihood volatility). Pooled OLS (circles) and within-image fixed effects (squares) both give $\hat{\beta}(C_{\text{res}}) > 0$ (destabilises) and $\hat{\beta}(C_{\text{prompt}}) < 0$ (stabilises). 1000-image 2B run gives larger magnitudes (+0.29/−0.32, $R^2 = 0.22$, $p < 10^{-130}$ / $p < 10^{-209}$).

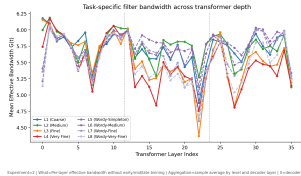


Figure 2: Layer-resolved bandwidth $G(t)$ across all 36 decoder attention layers (8B, 100 samples). Solid: primaries L1–L4. Dashed: wordy mirrors L5–L8. Vertical dashed lines mark layer-group thirds. The four stages of the trajectory are visible: a sharp layer-0 length prior (wordy below primary by −0.74), erasure by layer 2, length-insensitive content narrowing through mid, and a re-opened wordy-broader gap peaking at +0.33 at layer 31.

6 Discussion

The four-stage filter trajectory and the signature matrix together explain both empirical phenomena under a single mechanism. Fine reasoning is spectrally costly because the deep-decoder filter $W_t(\omega)^{(\text{last}_2)}$ concentrates downward onto DC-adjacent bins, where natural perturbations also place 80–96% of their energy; the resulting overlap $\langle W_t(\omega), \Delta F(\omega) \rangle$ grows, drift follows. Verbosity stabilises because the trajectory it induces—layer-0 length prior, mid-layer length-insensitive content narrowing, late-layer broadening—ends with $W_t(\omega)^{(\text{last}_2)}$ on a broader, less DC-concentrated shape than the terse counterpart, lowering the overlap and the predicted drift. The two effects look contradictory only if “complexity” is treated as a single axis: our mirror design separates them at the item level, and the FE regression confirms the opposite-signed split is a property of the model, not the corpus.

The mechanism produces a falsifiable prediction. If the C_{prompt} anchor is genuinely layer-0 in origin, then ablating or perturbing layer-0 cross-modal attention should preferentially dissolve the C_{prompt} effect while leaving the C_{sem}

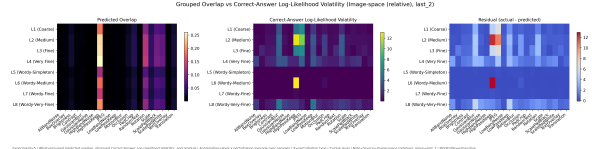
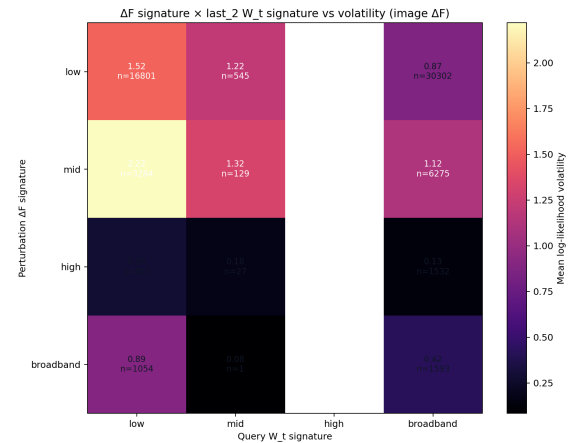


Figure 3: Overlap-law geometry (8B, last_2, image-space, target = loglik volatility). Left: predicted overlap $\langle W_t(\omega)^{(\text{last}_2)}, \Delta F(\omega) \rangle$ by (level $\times$ perturbation). Middle: observed volatility on the same grid. Right: residual after a single global slope. Hot spots in predicted line up with hot spots in observed. Grouped $r = 0.337$, per-sample $r = 0.315$; within-cell median $r = 0.543$ (2D, question-only).



Signatures use low/mid/high mass fractions plus normalized entropy. Correlations use $W_t \Delta F$ vs $\text{abs}(\loglik_drift)$ inside each cell.

Figure 4: Mean log-likelihood volatility by $(\Delta F, W_t(\omega))$ signature cell (8B, last_2, image-space, $n = 62,400$). In every ΔF row, the broadband- $W_t(\omega)$ column has the lowest mean drift: broadband filters mute drift across every perturbation profile. Column means: W_t -broadband 0.87 vs W_t -low 1.55 (44% reduction).

effect intact—a causal test purely behavioural accounts of prompt sensitivity cannot make. The practical payoff is a parameter-free deployment audit: estimate $W_t(\omega)$ from a handful of representative inputs, intersect with the corruption profile of interest, and read off $\langle W_t(\omega), \Delta F(\omega) \rangle$ as a calibrated fragility number. Prompt padding becomes a cheap, inference-time, training-free defence.

7 Limitations

The dual-force regression uses Qwen3-VL-2B; the headline overlap-law and signature-matrix numbers come from a 100-sample replication on Qwen3-VL-8B. Cross-model replication on LLaVA-NeXT and InternVL is in progress. C_{sem} is equal-weighted by design; the sign of every effect is invariant under any positive reweighting (App. B), but magnitudes

are not. $W_t(\omega)$ is estimated from attention, a cor-
relational lower bound on the true causal filter;
activation-patching remains future work. The mir-
ror construction is synthetic; human-judged para-
phrase mirrors are a natural extension. The pixel-
space radial Pearson–Spearman gap at last_2
(0.337 vs 0.186) indicates the linear correlation
captures part of its signal through magnitude scal-
ing on a heavy-tailed drift distribution; the 2D form
and vision-feature space close this gap.

8 Conclusion

We give a parameter-free, mechanistically
grounded predictor of vision–language model
fragility under image perturbation, derived from
cross-modal attention alone with no parameters
fit to behaviour. The predictor is item-level
(within-cell median Pearson $r = 0.543$ over
7,800 matched cells in which image, perturbation,
and severity are held identical and only the task
varies), recovers the population-level signal at
 $r = 0.337$, and is structurally validated by a
 $4 \times 4 \Delta F \times W_t(\omega)$ signature matrix in which
drift is largest where filter and perturbation
share a band, smallest where they don’t, and
broadband filters minimise drift across every
perturbation category. The mechanism explains
the verbosity-stabilises phenomenon (65–86%
drift-variance reduction in our mirror design),
grounds it in a four-stage layer trajectory of how
 $W_t(\omega)$ is built across decoder depth, and recovers
the underlying tug-of-war ($\tilde{\beta}(C_{\text{prompt}}) = -0.317$,
 $\tilde{\beta}(C_{\text{res}}) = +0.291$, $R^2 = 0.22$, $n = 8000$) in
one within-image fixed-effects regression. The
framework converts a black-box failure mode
into an engineerable, pre-computable quantity
and yields a free, training-free defence: pad the
prompt.

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A Theoretical Framework: Additional Details

Cross-modal attention to vision tokens. An image I is encoded into $N = H \times W$ vision tokens $\{v_{ij}\}$ on a rectangular grid; the cross-modal attention tensor $A_t(I) \in \mathbb{R}^{L \times H_{\text{heads}} \times T \times N}$ (decoder layers L , attention heads H_{heads} , T text tokens, N vision tokens), averaged over text tokens and heads, gives a spatial map $a_t^{(\ell)}(I)[i, j]$ per decoder layer (group). We bin radial frequencies into B linearly-spaced bands $\omega_1 < \dots < \omega_B$, $B = \lceil 0.6\sqrt{N} \rceil$ subject to a radial-annulus cap that drops empty bins ($B = 7$ on Qwen3-VL’s 12×16 patch grid; $B = 20$ on the historical 2B DC-ablation run).

IPR and shape diagnostics. The effective bandwidth $G(t) = (\sum_{\omega} W_t(\omega)(\omega)^2)^{-1}$ (eq. (1)) is in $[1, B]$ and quadratic, so bootstrap CIs are stable. To distinguish filter spread from filter location we additionally report *top-k mass* ($\sum_k \text{largest bands } W_t(\omega)$), *tail mass* ($\sum_{\omega \geq \omega_{B/2}} W_t(\omega)$), and the spectral *centroid* ($\sum_{\omega} \omega W_t(\omega)(\omega)$). Under late-layer downward concentration, top-2 mass rises with C_{sem} ($\rho = +0.6$) and tail mass falls ($\rho = -0.6$); $\rho(G(t)^{(\text{last-2})}, \text{rank}) = -1.0$ across L1–L4.

Cross-modal latent drift ΔZ . The mean ℓ_2 drift between clean and perturbed hidden-state trajectories at layer group ℓ is

$$\Delta Z(t, p)^{(\ell)} = \frac{1}{N^{(\ell)}} \sum_n \|z_n^{\text{pert}, (\ell)} - z_n^{\text{clean}, (\ell)}\|_2,$$

language tokens aligned positionally, vision-token spans aligned via patch-grid-aware resampling. $\Delta Z^{(\text{late})}$ tracks C_{sem} and $\Delta Z^{(\text{early})}$ tracks C_{prompt} , mirroring the behavioural split.

B Semantic Complexity Validity

$C_{\text{sem}} = E + A + R + O + D + \ln(1 + G_{\text{amb}})$ decomposes each question into the GOA program (Hudson and Manning, 2019; Johnson et al., 2017): entity references E , attribute predicates A , relations R , reasoning operations O , program depth D , plus a grounding-ambiguity term G_{amb} (number of scene-graph candidates satisfying the referring expression) from the referring-expression literature (Kazemzadeh et al., 2014; Mao et al., 2016). Each component weakly increases L1→L4 (e.g. entities $1 \rightarrow 2$, depth $1 \rightarrow 3$), so any positive-weight linear combination preserves $C_{\text{sem}}(\text{L1}) < C_{\text{sem}}(\text{L4})$ and all reported signs are weight-invariant; only magnitudes would

change under reweighting. Option hardness H_{opt} (same-category distractor count) enters eq. (5) as a separate multivariate control, not as a component of C_{sem} .

C Methodology: Configuration Detail

Perturbation bank. Geometric: cyclic horizontal translation, pad-or-crop, scale, scale-and-pad (black/white), rotation (6 ops; severity steps e.g. translation $\in \{4, 8, 12\}$ pixels). Occlusion: text overlay, box overlay, random-text overlay (3 ops). Spectral: low-pass keep, high-pass keep, low-band noise, high-band noise, all-band noise (5 ops; cut-off $\omega_c \in \{0.18, 0.28, 0.38\}$). For the 8B 100-sample run we use severity 1 only; for the 1000-image 2B run severities $\{1, 2, 3\}$ are pooled.

Model and extraction. Qwen3-VL-2B-Instruct (1000-image headline regression) and Qwen3-VL-8B-Instruct (100-sample overlap-law replication). Attention extracted at all 36 decoder layers via the eager attention implementation. Vision tokens identified via the special tokens `<vision_start>` / `<vision_end>`. Severity-1, no Hann tapering, DC retained, $B = 7$. The 8B run also stores 2D first-order overlap sidecars and a question-only filter sidecar (cross-attention computed from question tokens only, excluding answer-format prefix tokens).

Mirror prompt examples. L1 “Is there a cat in the image?” / L5 “Taking a careful look at the scene, can you tell me if there is a cat present?”. L4 “What is the colour of the cup that is on the table to the left of the chair?” (MCQ) / L8 “Taking your time to examine the scene, could you please tell me, of the listed options, what colour the cup is—specifically the cup that is positioned on the table to the left of the chair?” (MCQ, padded options).

D Task 1: Confirmatory Targets and Reconciliation

The accuracy-drop confirmatory regression (clean-correct → perturbed-wrong) recovers the same signs as the volatility regression but at $R^2 = 0.02$, an order of magnitude weaker. Thresholding to a binary $\{0, 1\}$ outcome discards perturbation-induced confidence displacements that do not cross the decision boundary; this is the source of the apparent “prompt-sensitivity” literature disagreement on signed complexity slopes. The 100-sample 8B replication under a cluster-robust (CR1) long-format specification gives $\beta_{C_{\text{prompt}}} = -0.054$

($p < 10^{-52}$) and $\beta_{C_{\text{res}}} = +0.035$ ($p = 1.1 \times 10^{-3}$, $n = 62,400, 100$ image clusters). The standardised $\tilde{\beta}$ in the main table and the raw CR1 betas are not directly comparable due to standardisation and the different n .

E Task 2: Per-Layer Bandwidth Tables

Layer-0 bandwidth across all 8 levels: $\{6.18, 6.00, 6.15, 5.74\}$ (L1–L4) vs $\{5.41, 5.21, 5.36, 5.14\}$ (L5–L8). Primary mean 6.02; wordy mean 5.28; gap -0.74 . By layer 2 the gap collapses to $|\Delta| < 0.10$.

Group $G(t)$	L1	L2	L3	L4	L5	L6	L7	L8
early	6.01	6.10	6.03	6.08	6.01	6.04	6.02	6.04
mid	5.86	5.89	5.64	5.49	5.90	5.85	5.77	5.51
late	5.99	5.88	5.74	5.56	6.01	5.96	5.84	5.74
last_2	5.93	5.92	5.72	5.68	5.94	5.98	5.91	5.96

Table 3: Layer-grouped bandwidth $G(t)$ for the 8 levels on the 100-sample 8B run. Mean wordy-vs-primary gap: early -0.03 , mid $+0.04$, late $+0.12$, last_2 $+0.15$.

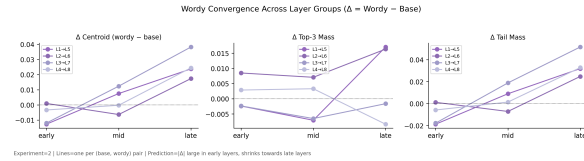


Figure 5: Per-layer wordy–primary bandwidth gap (8B). Strong negative gap at layer 0 (length prior); ≈ 0 through layers 1–12 and mid (13–23); growing positive gap through late (24–35), peaking at $+0.33$ at layer 31 (verbosity stabiliser engaged).

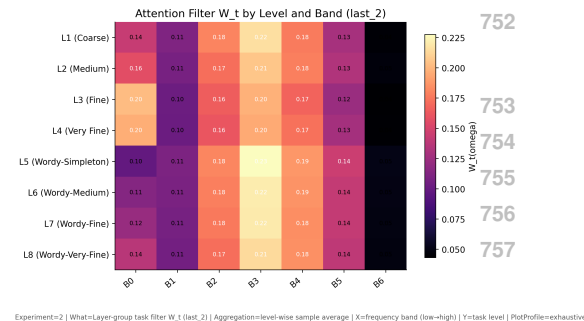


Figure 6: Cross-modal attention spectrum $W_t(\omega)$ across L1–L8 at $B = 7$ bins. Primaries (top) concentrate mass downward as C_{sem} rises; wordy mirrors (bottom) keep a broader distribution.

F Tasks 3–4: Internal Mechanism and Frequency Cutoff

Task 3 (layer-resolved correlation). Pearson grouped r between $\langle W_t(\omega)^{(\ell)}, \Delta F(\omega) \rangle$ and log-likelihood volatility at each layer group (8B, image-space, $n = 16,800$ grouped): overall 0.296, early 0.282, mid 0.281, late 0.306, last_2 **0.337**. 2D first-order is $\{0.279, 0.277, 0.261, 0.284, 0.315\}$. Vision-feature space, radial first-order: $\{0.218, 0.221, 0.228, 0.210, 0.197\}$ (last_2 weakest); 2D: $\{0.262, 0.243, 0.274, 0.277, 0.273\}$ (radial-reduction artefact).

Task 4 (cutoff law). Sweeping low-pass cutoff ω_c and defining the task-specific cutoff as the largest ω_c at which accuracy drops by $< \delta = 0.02$ gives ω_c falling monotonically with C_{sem} across L1–L8; wordy mirrors L5–L8 inherit the cutoff of their primary counterpart. Full per-level values in the released code artefact.

G Task 5: Full Overlap-Law Diagnostics

Spearman–Pearson gap. Pixel-space radial at last_2: $r = 0.337$, $\rho = 0.186$. Vision-feature radial: $r = 0.197$, $\rho = 0.197$ (gap closes). The pixel-space gap reflects heavy-tailed drift distribution; the 2D form and feature space remove it.

Matched-FE full table. Matched-FE Pearson on 7,800 matched (image $\times$ perturbation $\times$ severity) cells with all 8 levels per cell. Configurations: image-space last_2 radial $r_{\text{FE}} = 0.280$ (within-median 0.350); 2D 0.249 (0.457); 2D question-only 0.271 (**0.543**); vision-feature space radial -0.346 (-0.532); 2D $+0.193$ ($+0.016$).

Mirror split. Primaries-only grouped $r = 0.333$; wordy-only grouped $r = 0.318$ (last_2, radial). The law holds essentially equally on both halves of the design: verbosity damps the magnitude of drift without breaking its predictability.

DC handling. Paired 200-sample DC ablation on 2B with $B = 20$: $r = 0.471$ with DC retained, $r = 0.055$ with DC zeroed. Both $W_t(\omega)$ for fine tasks and $\Delta F(\omega)$ for natural perturbations concentrate 80–96% of their mass on DC-adjacent bins, so DC is load-bearing rather than cosmetic.

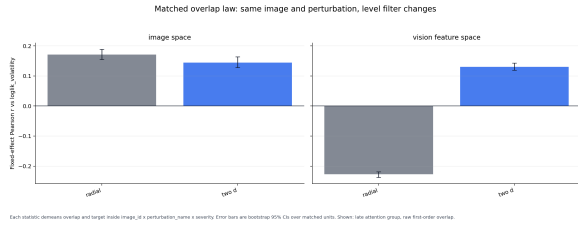


Figure 7: Matched-FE Pearson by domain and overlap form. Image-space: both radial and 2D positive (law works within-cell). Feature-space: radial flips negative (sign-flip), 2D recovers positive—the apparent flip is a radial-reduction artefact.

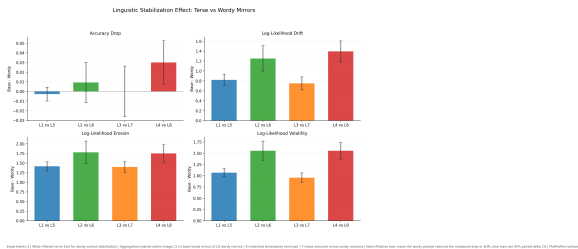


Figure 8: Linguistic stabilisation effect across four behavioural targets. Within-pair (base – wordy) for the four mirror pairs on accuracy drop, log-likelihood drift, log-likelihood erosion, and log-likelihood volatility. All 16 cells positive: every wordy mirror reduces every drift measure.

H Task 6: Full Signature Matrices

I Algorithm

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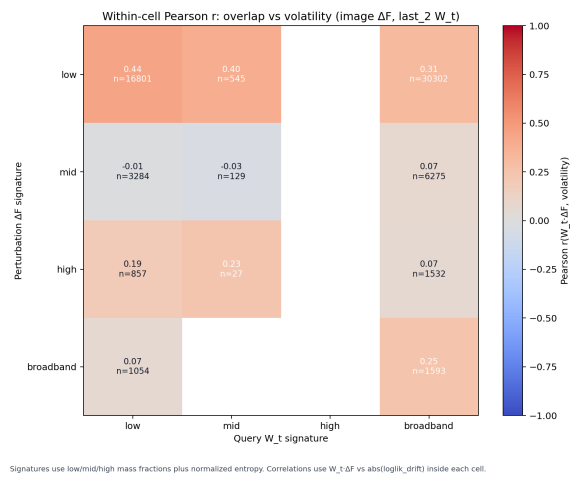


Figure 9: Within-cell Pearson r between predicted overlap and observed volatility by $(\Delta F, W_t(\omega))$ signature. Matched-filter peak at low $\times$ low diagonal ($r = 0.44$, $n = 16,801$); drops to $r = 0.31$ as $W_t(\omega)$ broadens away.

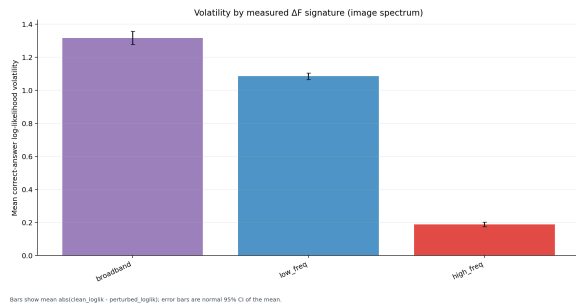


Figure 10: Mean log-likelihood volatility by measured ΔF signature alone. Broadband perturbations most damaging (1.32, $n = 14,384$); low-frequency next (1.09); high-frequency least (0.19), where $W_t(\omega)^{(last-2)}$ has thinned.

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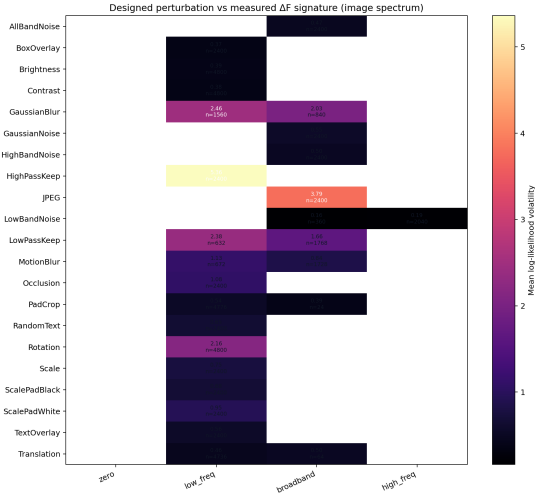


Figure 11: Designed perturbation operator vs measured ΔF signature. GaussianBlur, JPEG, LowPassKeep are broadband; HighPassKeep is low-freq (it removes low-freq, so the residual is low-freq); LowBandNoise is high-freq; occlusion / brightness / contrast are low-freq.

Algorithm 1 Spectral overlap predictor

Require: Image bank $\{I_i\}$, task t , perturbation p , VLM with attention hooks

- 1: **for** each image I_i **do**
- 2: Extract cross-modal attention $a_t(I_i)$ at last_2
- 3: 2D FFT, radial-bin $\rightarrow \hat{a}_t(I_i, \omega)$
- 4: Apply p to I_i ; compute $\Delta F(\omega)_p(\omega) = \mathbb{E}_I[|\mathcal{F}_2 I_p|^2 - |\mathcal{F}_2 I|^2]_\omega$
- 5: **end for**
- 6: $W_t(\omega) \leftarrow \text{normalize}(\mathbb{E}_i[\hat{a}_t(I_i, \omega)])$
- 7: $S_{\text{pred}}(t, p) \leftarrow \sum_\omega W_t(\omega)(\omega) \Delta F(\omega)_p(\omega)$
- 8: **return** S_{pred}

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